

Lazy Evaluation

CS 350

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April 10, 2025

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 - Thunks

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```
(define (subst [toReplace : Symbol]  
              [replacedBy : Exp]  
              [replaceIn : Exp]) : Exp ....)
```

Substituting Expressions vs. Values

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```
(define (interp expr)
  (type-case Exp interp
    ....
    [(appE fun arg)
     (type-case Value (interp fun)
       [(lamV var body)
        (let* ([argExp (Value->Exp (interp arg))])
          [subbedBody (subst var argExp body)])
        (interp subbedBody))]
       [else
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 - We don't

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- Works just as well, *but we get different behaviour*

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 - Lazy vs. Strict is one
 - We'll see more

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- If strict succeeds, lazy succeeds
- Sometimes, strict fails but lazy succeeds

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- Lets us define **infinite data structures** using recursion
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- See Racket examples in `lazy.rkt`

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 - This is basically just a lazy list, implemented similar way
- Can probably do something similar with iterators in C++

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- In strict language, has the same behaviour as the lazy version